

Chapter 14: Comparing Quantities

Worksheet 14: Comparing Quantities

Learning Objective: Apply percentage calculations to profit and loss, discount and marked price, and simple interest in real commercial contexts.

Worked Example:

A shopkeeper buys a fan for Rs. 1,200 and sells it for Rs. 1,440. Find the profit and the profit percentage.

Solution: Profit = $1,440 - 1,200 = \text{Rs. } 240$. Profit

Questions (1–20)

1. If the cost price is Rs. 500 and selling price is Rs. 600, find the profit.
2. Find 10% of Rs. 850.
3. A shirt marked at Rs. 1,000 is sold at a 20% discount. Find the discount amount.
4. What is the formula for simple interest?
5. If cost price = Rs. 800 and loss = Rs. 80, find the selling price.
6. A shopkeeper buys a toy for Rs. 250 and sells it for Rs. 300. Find the profit percentage.

-
7. A book marked at Rs. 450 is sold after a 10% discount. Find the selling price.

 8. Find the simple interest on Rs. 5,000 at 8% per annum for 2 years.

 9. An article bought for Rs. 1,500 is sold at a loss of 12%. Find the selling price.

 10. A shopkeeper marks an item 25% above the cost price of Rs. 800. Find the marked price.

 11. A trader buys a bicycle for Rs. 2,400 and sells it for Rs. 2,880. Find the profit percentage.

 12. A jacket marked at Rs. 1,600 is sold at a 15% discount. Find the discount amount and selling price.

-
13. Find the simple interest and total amount on a loan of Rs. 12,000 at 10% per annum for 3 years.
14. An item is sold for Rs. 920 after an 8% discount on the marked price. Find the marked price.
15. A shopkeeper sells two items at Rs. 500 each. On one he makes a 25% profit, and on the other a 20% loss. Find the cost price of each item, and the overall profit or loss.
16. A man invests Rs. 8,000 at 6% simple interest per annum. After how many years will the interest equal Rs. 2,400?
17. A shopkeeper buys 50 pens at Rs. 8 each. He sells 30 pens at Rs. 12 each and the rest at Rs. 6 each. Find his total profit or loss.
18. An article's marked price is Rs. 2,000 and is sold for Rs. 1,700, giving the shopkeeper a profit of about 13.33% on cost price. Find the approximate cost price and the discount percentage.

19. A shop advertises “Flat 30% off”. If the marked price is Rs. 1,500, calculate the price you would pay.
20. If you put Rs. 2,000 in a savings account giving 5% simple interest per year, calculate the interest earned in 3 years.

Reflection Question: Why do you think shops sometimes mark up prices before announcing a “discount”? What does this tell you about how to read sale advertisements?

Real-Life Application Question: Look at a bank or post office savings scheme interest rate (ask at home or check online or in a newspaper). Calculate the simple interest on Rs. 5,000 for 1 year at that rate.